

Azqa Jafar

AI & ML Engineer

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OBJECTIVE

Software Engineering graduate and AI/ML Engineer with hands-on experience building practical AI applications, LLM-powered systems, RAG pipelines, AI agents, backend APIs, and machine-learning workflows. Experienced with LangChain, LangGraph, MCP, vector databases, FastAPI, Python, and Flutter, with a focus on retrieval quality, reliable integrations, clean implementation, and user-friendly AI experiences. Comfortable working across the application lifecycle from data preparation and model integration to API development, testing, debugging, and deployment-oriented workflows. Interested in building production-focused AI products that solve real user and business problems.

CORE TECHNICAL SKILLS

Generative AI & LLMs: RAG, Agentic RAG, AI Agents, LLM Applications, Prompt Engineering, Context Engineering
Retrieval-Augmented Generation (RAG) & Retrieval: Semantic Search, Hybrid Retrieval, BM25, Vector Search, HNSW, Reranking, Embeddings, Chunking, Citation-Grounded Generation

Machine Learning: Supervised Learning, Unsupervised Learning, Classification, Regression, Feature Engineering, Model Training, Model Evaluation

Deep Learning: Neural Networks, CNNs, LSTMs, Transformers, Transfer Learning, Fine-Tuning, TensorFlow

AI Agents & Multi-Agent Systems: Autonomous AI Agents, Tool-Using Agents, Agentic Workflows, Multi-Agent Collaboration, Task Planning, Memory, Reasoning, Function Calling, LangGraph, CrewAI, MCP

Tools and Technologies: LangChain, LangGraph, LlamaIndex, CrewAI, MCP, Hugging Face, ChromaDB, FAISS, FastAPI, PostgreSQL, Docker, Git/GitHub, Postman, Jupyter Notebook, Kaggle, VS Code, OpenAI SDK

Basic Cybersecurity Concepts: CIA Triad, Authentication & Authorization, Access Control, Encryption Basics, Hashing, Network Security Fundamentals, Common Cyber Threats, Phishing, Malware, Vulnerability Basics

EXPERIENCE

AI Developer

12/2025 – Present

Minhaj Solution Software System

- Develop AI applications using Python, LLMs, RAG, and agentic workflows for practical business use cases, translating requirements into reliable AI-powered features.
- Build end-to-end RAG pipelines with document ingestion, chunking, embeddings, vector databases, semantic/hybrid retrieval, reranking, and context-aware response generation.
- Develop AI agents and multi-agent workflows with LangChain and LangGraph, including tool calling, task planning, memory, reasoning, state handling, and API integrations.
- Integrate MCP and external tools/services to extend agent capabilities and support structured tool-based workflows across AI applications.
- Design FastAPI and Flask backend services and REST APIs for AI/ML applications, frontend integration, and model-powered features.
- Work across ML/DL workflows including data preprocessing, model training, evaluation, integration, debugging, and performance-oriented iteration.
- Use ChromaDB, FAISS, PostgreSQL, Docker, Git/GitHub, and OpenAI-compatible SDKs to support production-oriented AI development and deployment workflows.

AI & Python Developer (Internship)

09/2025 – 11/2025

Efaida Technologies

- Developed and tested Python-based applications and backend features using clean, reusable code, structured modules, and practical software-development practices.
- Worked with AI/ML workflows including data preprocessing, model integration, REST APIs, databases, debugging, and Git/GitHub while supporting real application requirements.
- Assisted with LLM-based application features, API integration, testing, documentation, and issue resolution, gaining practical exposure to collaborative development and technical problem-solving.

PROJECTS

Enterprise AI Security Desktop Platform – Multi-Agent AI System

Developed a multi-agent AI security application for automated threat analysis, vulnerability assessment, secure workflow orchestration, and context-aware security insights. Designed agent-driven flows that combine RAG, MCP, vector retrieval, and external APIs to organize security context, support structured analysis, and generate clearer monitoring and reporting outputs. Built the solution with a Flutter desktop interface and Python/FastAPI services to connect AI reasoning with application workflows.

Tech: Python, FastAPI, LLMs, LangChain, LangGraph, RAG, MCP, Vector Databases, Flutter, Dart

AI-Powered System Monitoring Assistant — Flutter

Developed a Flutter-based AI monitoring assistant for tracking threads, application logs, and network activity in a unified interface. Integrated AI-assisted analysis to help interpret technical events, surface relevant information, and support faster diagnostics. Focused on clear presentation of monitoring data, API-driven integration, and a practical desktop experience for reviewing system activity.

Tech: Flutter, Dart, REST APIs, AI Integration, Log Analysis, Network Monitoring

ResearchLens AI — Evidence-Grounded AI Research Platform

Developed an evidence-grounded RAG application for analyzing research PDFs, performing semantic retrieval, and generating citation-backed answers with page-level source traceability. Implemented document ingestion, chunking, vector search, reranking, multi-provider LLM integration, authentication, conversation history, and source-aware response generation. Combined a React interface with FastAPI services and testing/deployment tooling to create a complete research-assistant workflow.

Tech: Python, FastAPI, React, RAG, ChromaDB, LangChain, LangGraph, SQLite, PyMuPDF, Groq, Claude, Docker, Pytest

UniGuide AI — AI-Powered University Assistant

Developed a RAG-based university assistant that provides context-aware responses to student questions about admissions, departments, courses, fees, and institutional policies. Structured university knowledge for retrieval, applied semantic search and LLM-based answer generation, and focused on grounding responses in relevant institutional information. Created the application as a practical student-support assistant with a simple interactive interface.

Tech: Python, NLP, RAG, LangChain, Streamlit

3D Multi-Plane Alzheimer's Disease Classification using Deep Learning (MPAFNet)

Final year project focused on early Alzheimer's disease classification from 3D multi-plane brain imaging data. Developed a custom MPAFNet-based deep-learning workflow to analyze complementary brain views, including preprocessing, feature extraction, transfer learning, training, and evaluation. The project explored how multi-plane information can strengthen representation learning and support reliable disease classification, while providing hands-on experience with medical-imaging AI and experimental model development.

EDUCATION

BS Software Engineering University of Okara | 2022 – 2026 | CGPA: 3.89/4.00

Graduated with a CGPA of 3.89/4.00 and strong academic performance. Built a solid foundation in software engineering, programming, databases, software quality, application development, AI/ML, and research-oriented problem solving. Academic and portfolio work included intelligent applications, backend services, deep-learning experimentation, and modern AI workflows.

Intermediate (ICS Physics) 2020 – 2022 | Marks: 950/1100

Completed Intermediate in Computer Science with Physics, Computer Science, and Mathematics, developing an early foundation in programming concepts, quantitative reasoning, computing fundamentals, and analytical problem-solving.

ACHIEVEMENTS

Prime Minister's Youth Laptop Scheme – Phase I

Selected as a recipient under the Prime Minister's Youth Laptop Scheme - Phase I on the basis of academic merit and eligibility, recognizing consistent academic performance during undergraduate studies.

Honhaar Merit-Based Graduation Scholarship

Awarded the Honhaar Merit-Based Graduation Scholarship in recognition of academic merit, punctuality, dedication, and sustained performance throughout the undergraduate degree.

PUBLICATIONS

Hierarchical Deep Learning with Stability-Driven Normalization and Regularization for Accurate Multi-Class Diabetic Retinopathy Diagnosis

Ain Shams Engineering Journal | Published 21 September 2026 | [ScienceDirect Publication Link](#)

Presented a hierarchical deep-learning framework for automated five-class diabetic retinopathy grading from retinal fundus images using the APTOS dataset. The approach combines hierarchical feature learning with stability-driven normalization and regularization to improve generalization across disease-severity classes. The study applies refined feature aggregation and systematic model evaluation to support robust multi-class classification. Experimental results report 98.91% test accuracy with near-perfect ROC-AUC performance across severity classes. The findings highlight the value of stability-focused deep-learning design for reliable computer-aided retinal screening.